

DEALER PROGRAMMING BOX FINAL REPORT

Project Engineer: Ben Lockwood

Project Title: BOSS Dealer Flash Tool

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GLOSSARY OF TERMS

Below is a list of technical terms used throughout the document to aid understanding of product functionality and standards.

CAN (Controller Area Network): Digital communication protocol used in vehicles and equipment to allow controllers and sensors to communicate.

TORO Diag Cable: USB-to-CAN adapter cable used to connect PC-based diagnostic software to BOSS modules.

BOB (Bluetooth On-Board): Bluetooth communication module used in newer systems.

VIM (Vehicle Interface Module): Connects vehicle controls to the BOSS control system.

M18: 18V lithium-ion battery platform by Milwaukee Tool.

IP44: Rating indicating protection from solid objects over 1mm and water splashing from any direction.

Key-Switch Simulation: A toggle or input used to mimic the ignition switch of a vehicle to allow diagnostics or updates to initiate.

The Dealer Programming Box fulfills a key service need for BOSS dealers by simplifying and accelerating firmware updates. By simulating a vehicle-side interface and providing safe, dual-mode power options, the device enables field and shop programming with minimal training. Development remains on schedule with full function expected by August 25, 2025.

1.0 INTRODUCTION

This final report summarizes the development of the Dealer Programming Box — a portable, dual-powered device for flashing and diagnosing BOSS modules. The project addresses a real-world need for dealer service efficiency and integrates CAN communication, power regulation, and hardware interfacing in a rugged enclosure. This report builds on the previously submitted Design Review and Scope documents and includes recent progress, testing strategies, and lessons learned during the co-op experience.

2.0 BACKGROUND

Currently, all BOSS firmware flashing and configuration updates require a fully installed system on a truck. This can be restrictive when trying to update spare parts, troubleshoot issues, or prep inventory. By providing a flash box that simulates a vehicle-side power and communication environment, dealers gain a streamlined solution that improves service turnaround and operational efficiency.

The Dealer Programming Box is designed to connect a PC (via TORO Diag USB-to-CAN cable) to any supported BOSS component using dedicated harnesses. It supports both shop (AC) and field (battery) power and includes features such as a key-switch toggle, blown-fuse indicator, and clean panel-mounted connections.

3.0 FUNCTIONAL OVERVIEW

3.1 System Architecture Overview

The Dealer Programming Box consists of a PC interface, internal CAN bus hub, multiple output harnesses, and dual-mode power regulation. The PC connects to the box using the TORO Diag USB-to-CAN cable. Inside the enclosure, the CAN signal is routed through an internal bus, with branches to three output ports and a bumper plug interface. Power regulation ensures a clean 12V output, and toggle switches control source selection and key-switch simulation.

A typical signal and power flow includes:

1. PC → USB-to-CAN → Internal CAN Bus
2. CAN Bus → Selected Harness (1, 2, or 3)
3. Battery or AC → Voltage Regulator → 12V Supply Line
4. Output routed to devices via appropriate connector (e.g., Display, Controller)

The internal layout separates low-voltage logic and CAN lines from higher-power distribution. Termination resistors are present within the board to meet ISO standards. LED indicators reflect the active power source and fuse integrity.

Inputs and Interface:

- USB to CAN via TORO Diag Cable (PC Connection)

- Toggle Switch (AC vs Battery Power)
- Toggle Switch (Key-Switch Simulation ON/OFF)
- Circuit Breaker or Inline Fuse with LED Fault Indicator

Outputs:

- Three dedicated harness ports:
 - Harness 1: Plow & Spreader Modules
 - Harness 2: SnowRator, BOB, VIM, Tri-Motor
 - Harness 3: Reserved for future expansions
- CFT Bumper Plug for prebuilt system interface

Power Supply:

- Milwaukee M18 Battery (field use)
- 120V AC Adapter (shop use)
- Regulated 12V DC output, compatible with all supported BOSS modules

Indicators:

- LED indicator for power mode
- LED for blown fuse

4.0 ENGINEERING SPECIFICATIONS

4.1 Electrical Design Details

The power system accepts either 120V AC from a wall outlet or 18V DC from a Milwaukee M18 battery. A toggle switch selects the active source. Power is stepped down and regulated to a stable 12V DC output. A circuit breaker (10A) or fuse protects the system, with an LED providing fault notification. Key power consumption specs:

- M18 Input Voltage Range: 16–20V DC
- Regulated Output: 12V $\pm$ 5%
- Maximum Load: ~5A (peak)
- CAN Bus Transceiver: MCP2562 or equivalent (ISO 11898-2 compliant)
- Fuse Type: Mini-blade, 10A, LED indicator socket

4.2 Enclosure Design Considerations

The enclosure is designed for shop and field durability. Ports are panel-mounted for ease of access. Layout priorities include:

- Top-mounted toggle switches (power source and key switch)
- Side-mounted barrel jack (AC in) and M18 dock
- Front panel: 3 harness ports + bumper plug + CAN input
- Laser-engraved or labeled indicators for clarity
- Optional carry handle and feet for stability on benches

Requirement	Description
Compatibility	Plow Modules, Spreader Modules, Displays, Hand Controllers, Tri-Motor, BOB, VIM
CAN Termination	Internal 120-ohm termination resistor (required per ISO 11898-2)
Operating Temperature	-20°C to 50°C

Power Modes	120V AC input or 18V Battery (switchable)
Protection	Inline fuse or breaker with LED indication
Environmental	Target IP44 protection (splash-resistant)
Portability	Hand-carriable, all connections internally mounted
Expansion	Harness 3 reserved for future module support

5.0 CURRENT STATUS

- CAN communication tested and verified with TORO Diag Cable
 - Harness pinouts defined and validated for Harnesses 1 and 2
 - Milwaukee M18 battery adapter tested successfully
 - Key-switch simulation and AC/DC switching circuit functional
 - Enclosure layout defined with spacing for all ports and toggles
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6.0 TIMELINE

Date Range	Task	
Phase	Dates	Status
Schematic + BOM Finalization	June 3–7	Completed
Enclosure + PCB Design	June 10–14	Completed
Prototype Assembly	June 17–21	Completed
CAN Comm Testing	June 24–28	Completed
Midpoint Review	July 1–3	Completed
Key Switch + LED Function	July 8–12	Completed
Final Assembly + Cable Routing	July 15–19	Completed
Internal Testing + Review	July 22–26	Completed

Mechanical Feedback + Rework	July 29–Aug 2	In Progress
Production Prep + Docs	Aug 5–Aug 16	Scheduled
Project Delivery	Aug 19–Aug 25	Upcoming

7.0 RISKS & MITIGATION

Risk	Mitigation Strategy
Power Loss During Flash	Stable voltage regulation and LED power indicators
Incorrect Harness Usage	Color-coded connectors, labeled ports
CAN Noise or Fault	Shielded cable, verified termination
Compatibility Drift	Harness 3 and modular interface allow expansion
Backfeeding to the AC/DC Power Supply	Adding a Diode to prevent backfeeding

8.0 CONCLUSION

The Dealer Programming Box fulfills a key service need for BOSS dealers by simplifying and accelerating firmware updates. By simulating a vehicle-side interface and providing safe, dual-mode power options, the device enables field and shop programming with minimal training. Development remains on schedule with full function expected by August 25, 2025.

9.0 FUTURE WORK & EXPANSION PLANS

Future enhancements to the Dealer Programming Box could include:

- Wireless (Bluetooth or Wi-Fi) programming via mobile app
- Integration with touchscreen or LCD for menu navigation
- CAN data logging for diagnostics and flash success verification
- Expanded enclosure model with modular plug slots for new hardware
- Automated VIN/module detection via stored module profiles

By planning for modularity and reserving internal space and interface pins, this project is positioned to scale with the evolving needs of BOSS product support.

10.0 Project Updates Since Design Review

Since the Design Review submission, the following tasks have been completed:

- **Schematics finalized:** All power, CAN, and switch circuitry is fully documented.
- **Circuit protection revised:** A **circuit breaker** with reset functionality replaced the fuse for better field usability.
- **Management review:** A formal DIC and schematic review meeting was held with BOSS engineering leadership. Feedback was incorporated.

- **Production prep:** Discussions and preliminary plans have started regarding sourcing, PCB finalization, and small-batch production.
- **Test Plan defined:** Criteria and procedures to verify functionality are now established and partially executed.

11.0 Test Plan.

Test	Method	Criteria	Status
Power Output	Measure 12V output under 5A load	12V $\pm$ 5%	Pass
CAN Communication	Connect to module via Diag cable	Confirm message exchange	Pass
Power Source Switching	Toggle between AC and M18	No brownout or reset	Pass
Key Switch Simulation	Monitor module response	Enters flash mode	Pass
Circuit Breaker Function	Trip via overload test	LED turns on, breaker resets	Pass
Harness Port Accuracy	Check pinout + continuity	Match DIC documentation	Pass

12.0 Lessons Learned

Technical

- CAN debugging required proper termination and shielding.
- Voltage regulation needs margin for startup transients.
- Circuit protection should consider technician usability — hence the switch to a circuit breaker.

Non-Technical

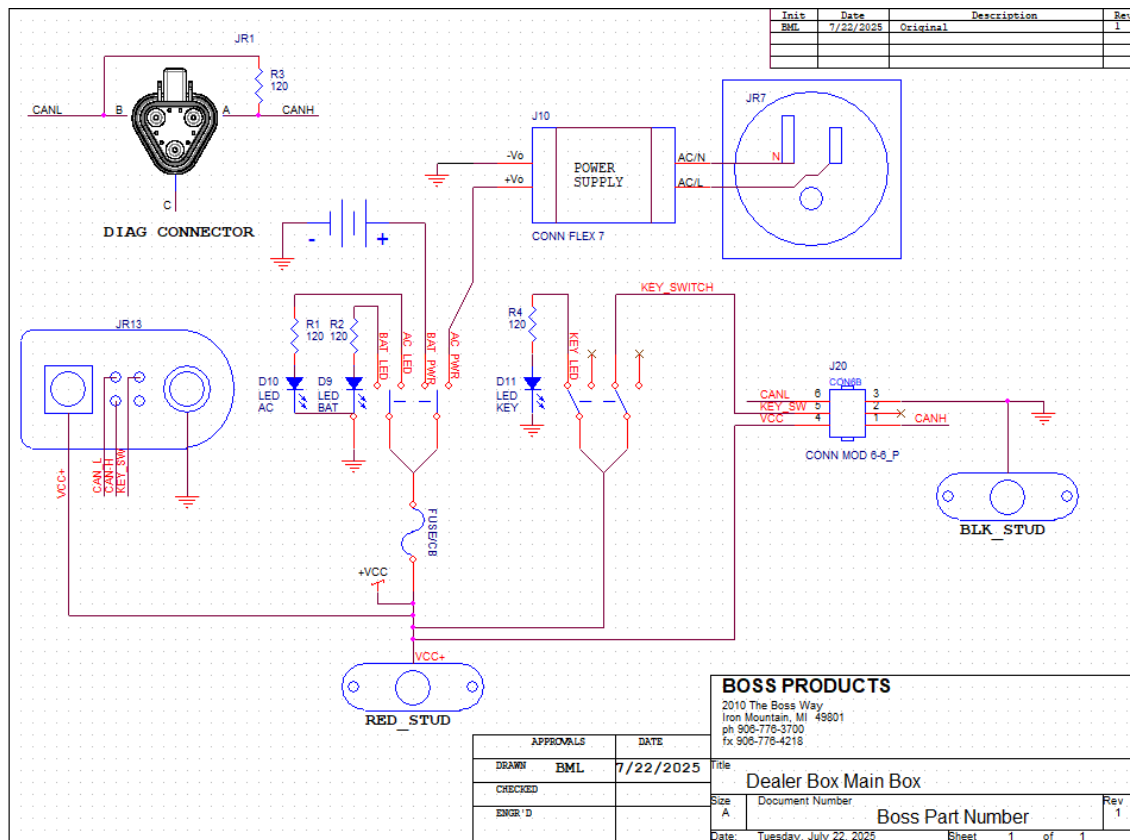
- Communication with hardware and mechanical teams was key.
 - Meet early and often, this helps so much to have weekly meetings to keep up with what management wants.
 - Keeping documentation up to date helped during meetings.
 - Prototype iteration was much faster once BOM and schematics were locked in early.
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Appendices

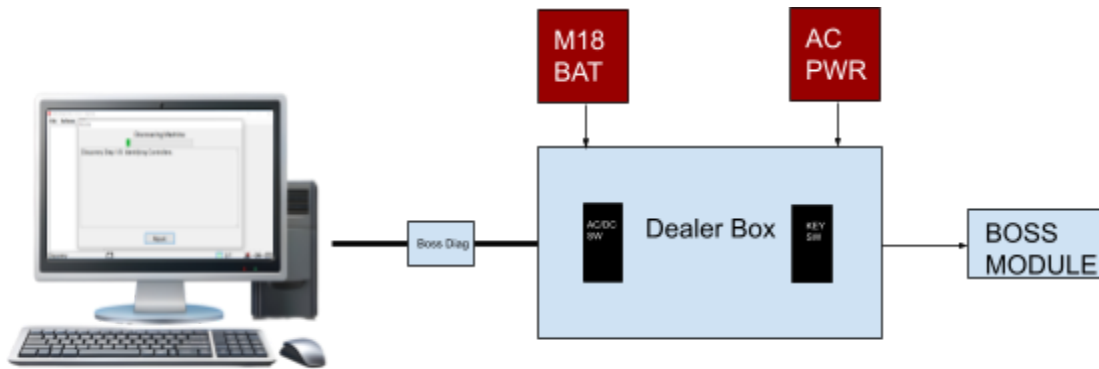
APPENDIX B: SCHEMATICS

This appendix will include full electrical schematics of the Dealer Programming Box. It will detail:

- Power input and switching circuitry (AC/Battery)
- Voltage regulation and fuse protection layout
- CAN transceiver circuit with termination
- Toggle switch and LED indicator integration



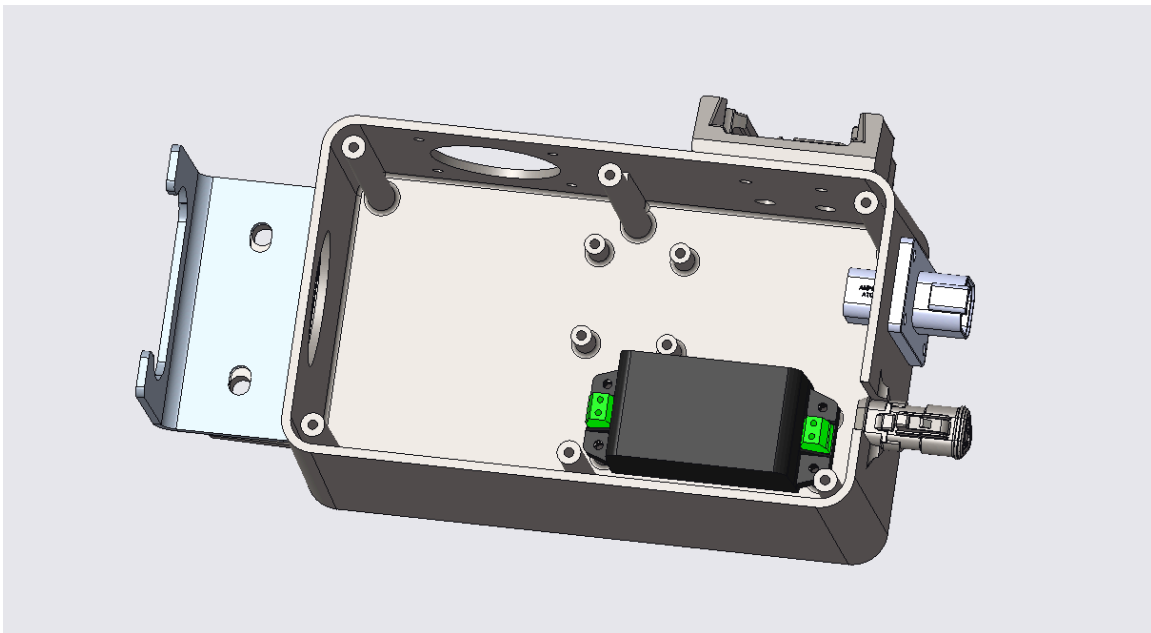
This has been approved and submitted with the DIC.

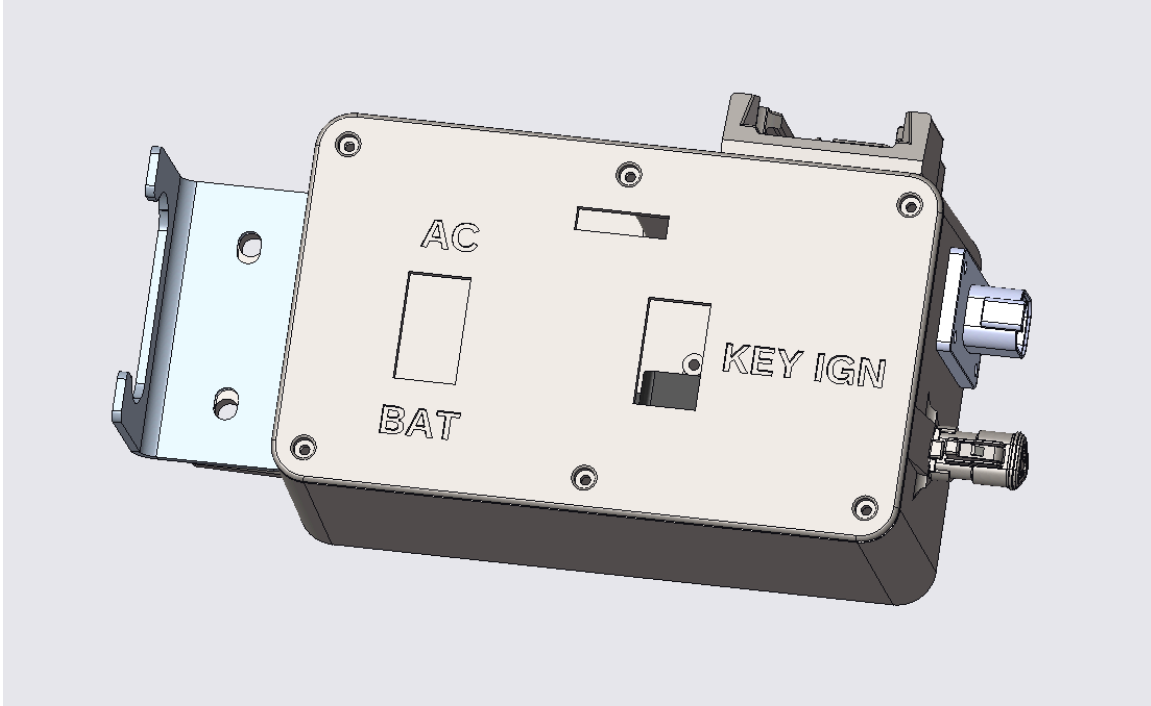


APPENDIX C: ENCLOSURE CAD MODELS

This section will contain mechanical design files including:

- Enclosure 3D model
- Front and top panel layouts
- Mounting hole patterns and cutout dimensions





CAD files will be released as part of the deliverables to BOSS Engineering upon project sign-off.

APPENDIX D: BILL OF MATERIALS (BOM)

The following table represents the initial Bill of Materials (BOM) for the Dealer Programming Box prototype. Values are subject to change as testing progresses:

Item No.	Component Description	Manufacturer	Quantity	Notes
1	Milwaukee M18 Battery Dock	Adapter / Generic	1	Powers system in field mode
2	AC-DC Adapter (120V to 12V)	Mean Well or Equivalent	1	For bench-top power

3	Toggle Switch – Power Select	DPDT - TBD	1	Selects between AC / Battery
4	Toggle Switch – Key Simulation	DPST - TBD	1	Simulates ignition/key signal
5	Circuit Breaker	OptiFuse	1	Shows C/B status visibly
6	CAN Transceiver IC	MCP2562 or TI SN65HVD230	1	Handles CAN communication
7	LED Indicators (Power/Fault)	Red/Green Panel Mount LEDs	2	Display system state
8	Enclosure (Plastic or Metal)	Custom / PolyCase	1	Includes panel cutouts
9	Harness Connectors (3 Sets)	Delphi / Molex	3	Harness 1, 2, and future
10	USB to CAN Adapter (TORO Diag Cable)	Existing Equipment	1	External PC interface

This list will be expanded in the final design to include wire gauges, PCB components, fasteners, and labels.

Appendix E: Hand Calculations

E.1 – Power Consumption

To estimate total power draw during module flashing and diagnostics:

$$P = V \times I = 12V \times 5A = 60W^*$$

This value represents the peak expected power load based on the maximum combined current from connected modules and indicator LEDs.

E.2 – Fuse Sizing

The box includes a 10A mini-blade fuse with an LED indicator. Fuse sizing is calculated to allow peak currents without nuisance trips, while still protecting wiring and components:

$$\text{Required fuse rating} = 2 \times I_{\text{peak}} = 2 \times 5A = 10A^*$$

A 10A slow-blow fuse was selected to accommodate inrush currents.

E.3 – Voltage Drop in Harnesses

Using 16 AWG copper wire (typical resistance $\approx 0.004 \Omega/\text{ft}$), and estimating a 5-foot round-trip run (10 feet total), with 5A current:

$$V = I \times R = 5A \times (0.004\Omega/\text{ft} \times 10\text{ft}) = 0.2V^*$$

Resulting voltage at the load = $12.0V - 0.2V = 11.8V$, which is acceptable for all supported modules.

E.4 – Battery Runtime (M18™ FORGE™ HD12.0)

The Milwaukee M18 RedLithium FORGE battery (HD12.0) provides:

$$\text{Energy} = 18V \times 12.0Ah = 216Wh$$

$$\text{Runtime} = 216Wh / \text{Load Power (W)}$$

Load Condition	Power (W)	Runtime (hours)
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Full Load (flashing)	60	3.6
Moderate Load (mixed use)	40	5.4
Idle / LEDs only	20	10.8

E.5 – CAN Bus Termination Resistance

Standard CAN bus requires 120Ω resistors at each end of the network.

$$R_{total} = 1 / (1/120) + (1/120) = 60\Omega$$

An internal 120Ω resistor is included in the circuit, and the second termination is assumed to be present at the connected module, satisfying ISO 11898-2 compliance.

* = Note these are approximate values, real values will be added when I finish measurements